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Abundance and distribution of pelagic piscivorous fishes in the Columbia River plume during spring/early summer 1998–2003: Relationship to oceanographic conditions, forage fishes, and juvenile salmonids

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Abstract

From 1998 to 2003, we observed large fluctuations in the abundance and distribution of four pelagic predatory (piscivorous) fishes off northern Oregon and southern Washington, USA. Fluctuations in predatory fish species composition and abundance were strongly linked to the date of the spring transition and to ocean temperatures. Predatory fishes, forage fishes, and juvenile salmonids had distinct spatial distributions, with predators distributed primarily offshore and forage fish and salmonids onshore, but this varied depending on ocean conditions. We suggest that predatory and forage fish distributions respond to ocean temperatures, predator/prey interactions, and possibly turbidity. A shift in ocean conditions in 1999 decreased overall predator fish abundance in the Columbia River plume, particularly for Pacific hake. Marine survival of juvenile salmon started to increase in 1999, and forage fish densities increased in 2000, lagging by one year.
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Keywords: California current; Columbia river plume; Predators; Pacific hake; Forage fish; Salmonids

1. Introduction

Predation influences ecosystem structure (Pace, Cole, Carpenter, & Kitchell, 1999; Shears & Babcock, 2003), the abundance of fish stocks (Bax, 1998; Borgstad & Mehl, 1997; Tsou & Collie, 2001) and is an important parameter to include in fish stock assessments (Hollowed, Ianelli, & Livingston, 2000). For example,

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Livingston and Jurado-Molina (2000) found that fish predation influenced recruitment of walleye pollock (*Theragra chalcogramma*) in the Bearing Sea. In the Barents Sea, pelagic fish stocks fluctuate depending on shifting predation pressures (Gjøsaeter, 1995). The abundance of capelin (*Mallotus villosus*) stocks in the Northwest Atlantic also appears to be driven by changes in predation pressure (Carscadden, Frank, & Leggett, 2001). Swain and Sinclair (2000) found evidence that predation by Atlantic herring (*Clupea harengus*) and mackerel (*Scomber scombrus*) affected the recruitment of cod (*Gadus morhua*) in the southern Gulf of St. Lawrence.

Pacific hake (*Merluccius productus*) (also known as Pacific whiting) is the most abundant predatory (piscivorous) fish on the west coast of North America (Methot & Dorn, 1995). Predation by Pacific hake affects the abundance of Pacific herring (*Clupea pallasi*) off Vancouver Island, Canada (Ware & McFarlane, 1995) and pink shrimp (*Pandalus jordani*) off Oregon and Washington (Hannah, 1995). Pacific hake have also been found to be major predators of juvenile salmon in Alberni Inlet and Barkley Sound, Canada (Beacham, 1991).

Pacific hake migrate north in the spring/summer to feed off Oregon/Washington/British Columbia, and south in fall and winter to spawn off California (Dorn, 1995). During spring, hake are generally found over the continental slope, but are reported to move onto the shelf in June (Methot & Dorn, 1995). Hake show an age/size specific migrational pattern, with larger/older, usually female hake, migrating the farthest north (Dorn, 1995; Methot & Dorn, 1995; Ware & McFarlane, 1995) and smaller, younger hake, migrating the least (Dorn, 1995). Northern and southern migrations appear to be strongly affected by ocean temperatures (Dorn, 1995) and perhaps food resources (Benson, McFarlane, Allen, & Dower, 2002), with warm or cold spring temperatures, respectively, increasing or decreasing the number of hake migrating to Canadian waters (Dorn, 1995; Ware & McFarlane, 1995). There is also evidence that when winter ocean temperatures are warm, many Pacific hake do not migrate south, but over winter and spawn off the Pacific Northwest and Canada (Benson et al., 2002; Hollowed, 1992).

Other important coastal pelagic predatory fishes off Oregon/Washington include jack mackerel (*Trachurus symmetricus*), chub mackerel (*Scomber japonicus*), spiny dogfish (*Squalus acanthias*), and adult salmonids (Brodeur & Pearcy, 1986). Very little is known about the migrational movement of chub and jack mackerel, except that they generally arrive in Pacific Northwest coastal waters in mid-summer (June; MacCall et al., 1980) and are more abundant during warm ocean years, such as El Niño years (Brodeur & Pearcy, 1986; Pearcy & Schoener, 1987). While no directed fishery for jack mackerel presently exists in the Pacific Northwest, it is often abundant and the primary bycatch in the Pacific hake fishery (Wiedoff, Conrad, & Parker, 2003). Spiny dogfish has a nearly worldwide distribution (Compagno, 1984), and there is a Northeast Pacific population that ranges from northern California to Alaska (McFarlane & King, 2003). The once-large Northeast Pacific fishery for spiny dogfish peaked in the 1940s (Hart, 1973) but is relatively small now because of limited demand (PSMFC, 2001). Spiny dogfish was harvested in the 1940s for its liver, which had high concentrations of vitamin A, but the development of synthetic vitamin A eliminated this demand.

There are primarily four species of forage fish that reside in the coastal zone off the Columbia River and the Pacific Northwest: northern anchovy (*Engraulis mordax*), Pacific herring, whitebait smelt (*Allosmerus elongatus*), and Pacific sardine (*Sardinops sagax*). Until the mid-1990s, Pacific sardine were not found off the Pacific Northwest, but they now support an important fishery (Emmett et al., in press). These forage fishes are important prey for many species of large fish, seabirds, and marine mammals, but relatively little information has been published on their abundance and spatial/temporal distributions off Oregon/Washington (Brodeur, Fisher, Emmett, Morgan, & Casillas, 2005; Brodeur & Pearcy, 1986; Emmett & Brodeur, 2000).

There were large changes in abundances of pelagic predatory and forage fishes off Oregon and Washington during the 1980–1990s and concurrent fluctuations in marine survival of salmon (Emmett & Brodeur, 2000). Starting in 1999, large increases in marine survival of salmon led to rebounding salmon stocks in the Northwest (Williams et al., 2005). This change in marine survival occurred simultaneously with a large shift in oceanographic conditions in the California Current ecosystem (CCE) (Peterson & Schwing, 2003; Schwing et al., 2002). Were there changes in the distribution and abundance of pelagic predators in the CCE as a result of this shift? If so, could these changes be responsible for the observed increase in marine survival of salmon? Here we identify changes in the abundance and distribution of the four most abundant pelagic predatory fishes off the Pacific Northwest from 1998 to 2003 and characterize the associated oceanographic conditions during

this time. We also begin to identify whether there is any relationship between predatory fish populations, forage fish populations and juvenile salmonid marine survival.

2. Methods

2.1. Study area

The study area was located off the mouth of the Columbia River (Fig. 1) and in the northern portion of the CCE. This portion of the CCE has seasonally variable winds and currents, with northwest winds, southerly currents (California Current) and upwelling in the summer, and south winds, northerly flows (Davidson Current), and downwelling in winter (Hickey, 1989; Hickey & Banas, 2003). Upwelling strongly affects primary productivity and oceanographic conditions in the study region but is highly variable. The effectiveness of northerly winds in generating it, and the amount of deep cold water actually rising to the surface, can be limited if the thermocline is deep. The date of the spring transition, when upwelling begins, sea level drops, and nearshore surface currents become southerly, is an important parameter determining coastal ocean environmental conditions and ocean survival of coho salmon (*Oncorhynchus kisutch*) (Logerwell, Mantua, Lawson, Francis, & Agostini, 2003). The low-salinity Columbia River plume is a dominant feature of the study area and is usually located on the continental shelf (nearshore) off Washington during winter and beyond the shelf off Oregon during summer (Hickey & Banas, 2003). Columbia River flows are highly regulated by dams, with peak flows much lower than historical levels (Sherwood, Jay, Harvey, Hamilton, & Simenstad, 1990); highest flows are generally in May/June and lowest flows in August/September.

Ocean temperatures, salinities, and densities at 50-m are good indicators of nutrient rich source water that surfaces when upwelling occurs. This water lies below the Columbia River plume and is influenced by upwelling intensity and large-scale fluctuations in the California Current (Hickey & Banas, 2003). In 2002, these deep

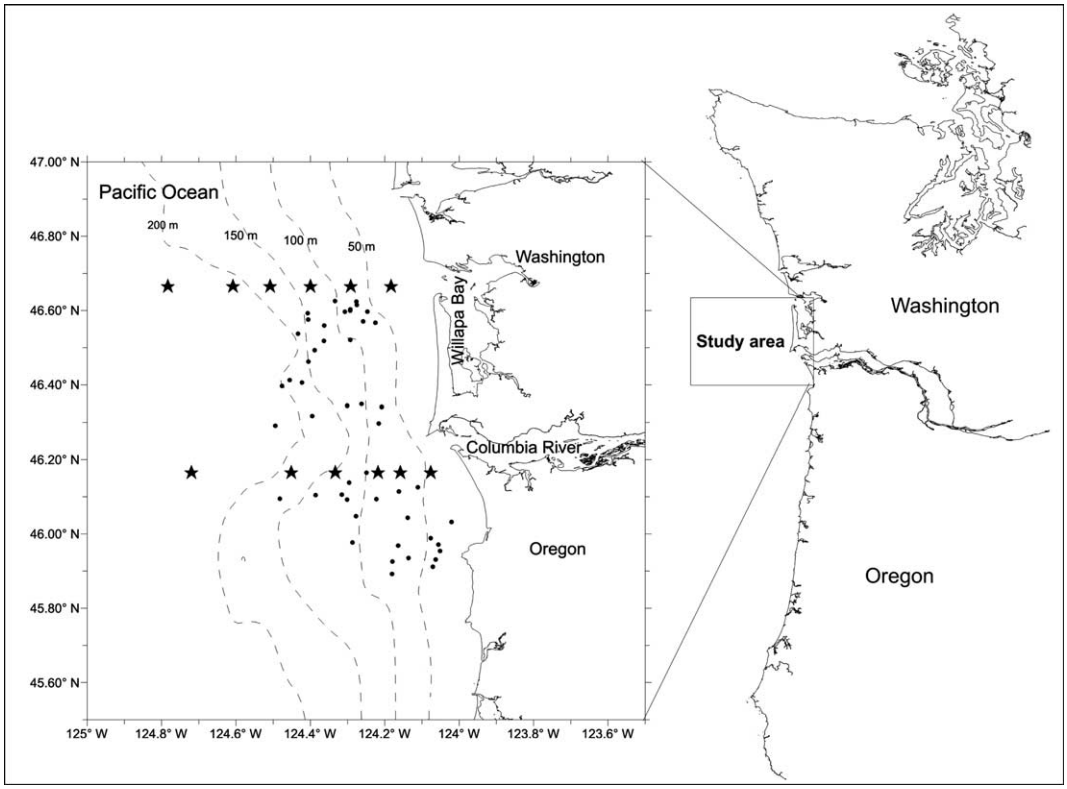


Fig. 1. Location of surface trawl stations sampled during spring/early summer 1998 (●) and 1999–2003 (★) off the mouth of the Columbia River.

waters saw an intrusion of deep, cold, low-oxygen Subarctic water that originated in the North Pacific (Kosro, 2003; Strub & James, 2003). The influx of this nutrient rich water increased primary productivity (Wheeler, Huyer, & Fleischbein, 2003), reduced dissolved oxygen levels, and caused mortality of bottom-dwelling fishes and crabs off central Oregon (Grantham et al., 2004), and probably modified the movements of pelagic fishes.

2.2. Sampling

We conducted our study from 1998 to 2003 and will refer to it as the Predator Study. The first year (1998) was used principally to identify the appropriate gear type and sampling scheme to collect pelagic fishes near the surface. As a result, many different stations were sampled in 1998 (Fig. 1). During all years we sampled primarily at night to take advantage of the diel vertically migratory behavior of pelagic fishes (Bailey, Francis, & Stevens, 1982; Blaxter & Holliday, 1963) and because surface trawls are more effective at night due to less visual avoidance (Dotson & Griffith, 1996).

In 1998 the fishing gear was deployed one or two times each night when the depth sounder indicated fish (undetermined species) were around the vessel and near the surface, as one of the initial study objectives was to identify whether our surface trawl gear fished effectively. Blind tows (no prior information about fishing conditions) were also conducted. We report 1998 data only from collections when a rope trawl was used (1 June 1998-on), as this was the fishing gear finally identified as most effective. Because the 1998 sampling scheme differed from that in 1999–2003 (different stations), calculated fish densities in 1998 may not be completely comparable to later years. While we did not have any May 1998 surface-trawl catch information, our subsurface trawl catches indicate that hake were in the study area (unpublished data). As such, fish densities were probably similar to what we observed in June 1998. Fish catches and oceanographic conditions differed widely between 1998 and 1999–2002. The 1998 data thus provided valuable contrasting information on fish/oceanographic relationships that we believe are important to include in this analysis. Finally, fish catch data from 2003 indicated that 1998 data were not aberrant.

From 1999 to 2003 we sampled at pre-determined stations along two transect lines north and south of the mouth of the Columbia River (Fig. 1). Six stations were sampled along each transect, with the first station as close to shore as possible (30-m deep), and the farthest station approximately 55.6 km from shore. Sampling was conducted at night, approximately every 10 days from mid-April to July-early August from 1998 to 2003, for a total of 20 sampling days (10 sampling cruises per year). From each trawl, all fish species were identified, enumerated and measured, except when large catches occurred. For each large haul, a random sample of 30 individual fish from each species was measured, and a subsample (approximately 5–30 kg, depending on fish size) of each species was counted and weighed, and the remaining mass weighed. Total numbers of each species captured were then determined by adding the number in the subsample to the number calculated by dividing the mass weight of the species by the average weight/individual calculated from the subsample.

All salmon were separated into age classes, subyearling (0.0-age), yearling (1.0-age) and ($\geqslant$2-year-old-age) by length. Size of each age group varied monthly, so age was identified by length-frequency analysis and historical salmon size at age information (J. Fisher, Oregon State University, Corvallis, OR, pers. comm.). We used the salmon age nomenclature of Koo (1962) with number before the period indicating winters spent in freshwater after hatching and before migrating to sea, and the number after the period indicating winters spent in the ocean.

From 12 June 1998 to 2003, all samples were conducted with one gear type aboard a chartered commercial trawler. All surface trawling was conducted with a 264-rope trawl with 3-m foam-filled Lite doors, designed and built by NET Systems.[1] This gear is also used to capture juvenile salmonids and associated fishes off southeast Alaska (Murphy, Brase, & Orsi, 1999) and California (MacFarlane & Norton, 2002). The trawl is 100-m long with a fishing mouth area 28-m wide and 12-m deep. The effective mouth area (336 m^2) was identified in earlier work using a backward-looking net sounder (Emmett, Brodeur, & Orton, 2004). The net was fished close to the surface, but the head rope depth was usually 1–1.5 m (Krutzikowsky & Emmett, 2005). Mesh size ranged from 126.2 cm in the throat of the net near the jib lines to 8.9 cm in the cod end. A 6.1-m long, 0.8-cm stretch knotless

[1] Mention of trade names does not mean endorsement by NOAA, NMFS, or Department of Commerce.

web liner was sewn into the cod end to capture small fishes and invertebrates. The 264-rope trawl was fished by towing it 137 m behind the vessel traveling approximately 2.9 knots (1.5 m s^{-1}) for 30 min. However, starting in 2001, trawl times were shortened to 15 min because of very large catches of forage fishes. Fish catches were standardized by calculating densities [number $(10^6\ m^3)^{-1}$] using the distance fished (geographic distance between beginning and end of trawl) multiplied by the effective mouth area.

Estimates of predator densities presented here must be considered as minima, particularly for very large predators such as adult salmon and sharks. Large predatory fishes are fast swimmers that may evade trawl nets, and generally do so during daylight. For example, although the Columbia River has relatively large runs of adult salmonids, we captured very few. On one occasion we placed a video camera on the head rope during daylight trawling operations and saw large adult salmonids both enter and leave the net opening. Evidently adult salmonids can easily evade a surface trawl traveling at 1.5 m s^{-1}. Better estimates of very large and fast predatory fishes, such as adult salmonids, would require different sampling gear. Nevertheless, our nighttime surface trawling appeared to be effective at capturing Pacific hake, a relatively slow swimmer and mackerel, a medium-speed swimmer. This rope trawl was originally designed to capture fishes, such as hake, at mid-water depths but modified (foam filled doors and floats attached to head rope) to fish at the surface (Steve Patterson, Noreastern Trawl Systems, Bainbridge Island, WA, pers. comm.). Mid-water trawls are also used to capture jack mackerel commercially (Arcos, Cubillos, & Núñez, 2001). Our catches of large dogfish shark may not be quantitative, but are presented because they were commonly captured and we believe the data does represent relative abundance differences.

Our night-time surface trawling appears to be effective for forage fishes (Emmett et al., 2004; Krutzikowsky & Emmett, 2005). Because of their small size (most averaged <200 mm FL) and slower swimming speeds, forage fishes are easily captured by the net. The efficiency of the trawl for forage fish was verified by large catches once they became abundant. For example, we also caught large numbers of Pacific sardine/trawl when the commercial sardine purse-seine fishers did.

Juvenile salmon are similar in size to forage fishes, and the surface trawl effectively captures juvenile salmon, especially during the day (Murphy et al., 1999; Orsi, Sturdevant, Murphy, Mortesen, & Wing, 2000). The surface trawl is less effective at capturing juvenile salmon during night, possibly because of diel behavior (Krutzikowsky & Emmett, 2005). However, the annual juvenile salmon density estimates provide consistent relative densities.

Annual trawl surveys could not be conducted on exactly the same dates each year due to weather and other factors. To allow annual comparisons, data were grouped by proximate calendar dates. Dates of each sampling effort and how the data were grouped are shown in Table 1.

To identify the overlap in habitat use between predators, forage fishes, and juvenile salmonids, their distributions and species densities were compared as a function of distance from shore. Since nighttime sampling may under represent juvenile salmonid densities (Krutzikowsky & Emmett, 2005), distributions of juvenile salmonids off Oregon and Washington from the NOAA Fisheries/Bonneville Power Administration (BPA) Columbia River Plume Study (June 1998–2003) are presented. The BPA Plume Study used the same surface

Table 1
Dates of sampling each year and the grouping dates used to show seasonal differences in abundance

Grouping date	Year					
	1998	1999	2000	2001	2002	2003
15 Apr		15-Apr				
23 Apr		23-Apr		26-Apr	24-Apr	24-Apr
5 May		5-May	30-Apr	8-May	3-May	3-May
14 May		14-May	11-May, 18-May	18-May	14-May	14-May
28 May	1-Jun	28-May	30-May	29-May	24-May	24-May, 5 June
13 Jun	13-Jun	13-Jun	14-Jun	8-Jun	15-Jun	14-Jun
26 Jun	28-Jun	26-Jun	29-Jun	30-Jun	27-Jun	27-Jun
7 Jul		7-Jul	11-Jul	11-Jul	10-Jul	10-Jul
15 Jul	13-Jul	15-Jul	22-Jul	20-Jul	23-Jul	23-Jul
28 Jul	30-Jul	28-Jul		31-Jul	2-Aug	2-Aug
28 Jul	12-Aug					

trawl and protocol, but sampled during daylight hours on a series of transects ranging from Cape Flattery, WA to Newport, OR (Fig. 2) (Emmett & Brodeur, 2000).

2.3. Physical measurements

Temperature and salinity profiles were collected at all trawl stations (from this study and the BPA Plume Study) by lowering a SeaBird SB-19 conductivity, temperature, and depth (CTD) probe to 100 m depth (or just off the bottom). Starting in 2002, a transmissiometer was attached to the CTD used in the BPA Plume Study to measure water clarity. Upwelling values and their anomalies at 45°N 125°W were obtained from NOAA/NMFS Pacific Fisheries Environmental Laboratory web site (http://www.pfeg.noaa.gov/). Annual dates for the spring transition, defined by when northwest winds, upwelling and equator-ward currents begin seasonally, were identified by examining changes in daily plots of Oregon/Washington coastal upwelling winds and sea level height filtered at 10 and 90 days, (Logerwell et al., 2003; E. Logerwell, NOAA Fisheries, Seattle, WA, pers. comm.). Columbia River flows, which influence surface salinities within the study area, were obtained from US Geological Survey Beaver Army Terminal located near Quincy, Oregon (46°10′54″N, 123°10′58″W) (http://waterdata.usgs.gov/nwis/nwis).

2.4. Statistical analysis

A Kruskal–Wallis test was used to identify statistical differences in physical oceanographic features (Columbia River flows, ocean temperatures, ocean salinities, etc.) and fish densities (by species and combined) between years because these data were not normally distributed. Dunn's multiple comparison test identified which years were significantly different when a Kruskal–Wallis test identified differences among years. A

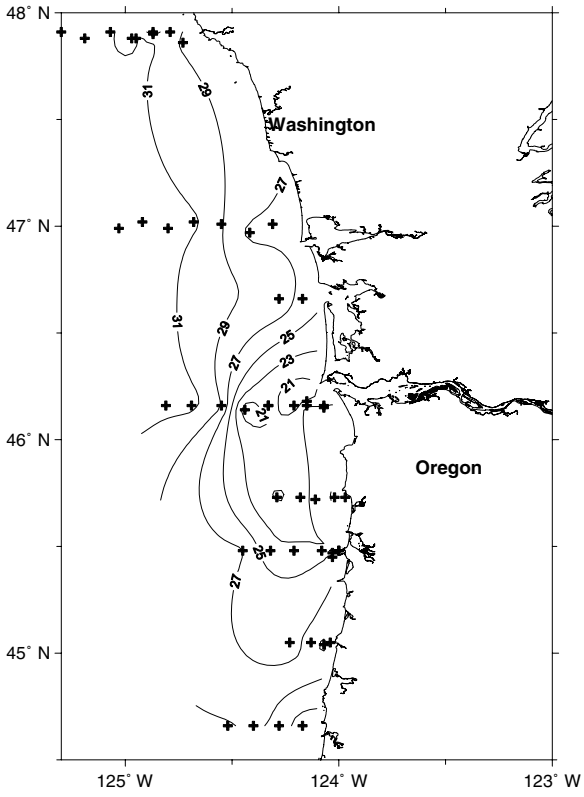


Fig. 2. Location (+) of daytime surface trawl stations sampled during June 1999 by BPA Plume Study. Surface salinity isopycnals are also shown.

2×2 χ^2 contingency test with Yates' correction was used to evaluate the relationship between the presence of predatory fishes and forage fishes and juvenile salmonids in the surface trawls.

A general linear model (GLM) was used to identify which physical and biological factors were related to the annual variation in average predatory/forage fish densities (all species combined and individually). Average fish densities were log transformed before analysis. Residuals were tested for Normality using the χ^2 goodness-of-fit statistic. Physical factors in the analysis included: date of the spring transition, Columbia River flow, ocean temperature, salinity, density, and distance from shore. Spring transition was quantified in Julian days. A sum of squares F test was used to identify the relationship between species densities and distance from shore after accounting for the effect of year (Ramsey & Schafer, 1997). We considered a $p < 0.05$ to indicate a significant relationship for all statistical tests.

GLM was also used to identify the relationship between average May/June predator/forage fish densities and coho salmon marine survival in the Oregon Production Index (OPI) area (PMFC, 2005). Coho salmon OPI survival was calculated by dividing coho salmon adult and jack returns by number of smolts released.

3. Results

3.1. Physical oceanographic conditions

Sea surface salinities and temperatures in our study area were strongly affected by Columbia River flow. For example, in June 1999, a month and year with average Columbia River flows (approximately 10,000 $m^3 s^{-1}$; Sherwood et al., 1990), the Plume Study documented a low-salinity plume that spread both north and south along the Oregon and Washington coast (Fig. 2). The depth of this low-salinity/low-density water was generally <20 m (Fig. 3) and depended not only on the amount of freshwater flow, but general oceanic conditions at that

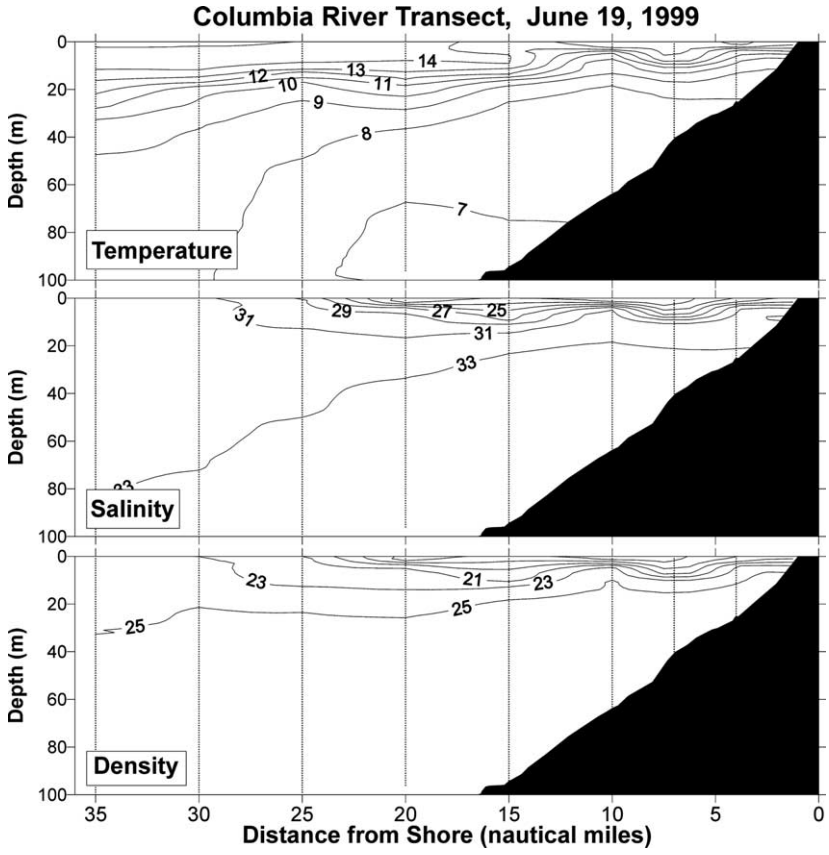


Fig. 3. Onshore–offshore sections of temperature, salinity, and density just south of the Columbia River, 19 June 1999.

time (winds and currents). During 2001, a drought year, average Columbia River flow was about half the average during the other five years. Annual and monthly Columbia River flow varied significantly (Kruskal–Wallace, $p < 0.0001$), especially during the spring (April–June) (Fig. 4).

As expected, the seasonal pattern of northerly winds and annual upwelling showed a general increase from April to July and August (Fig. 5A). While we identified no annual upwelling differences from 1998 to 2003 (Kruskal–Wallace, $p = 0.504$; overall median upwelling from April–August was 34 $m^3/s/1000$ m of coastline), there were significant interannual differences within months (Table 2). For example, in April 2003 median upwelling was strongly negative (downwelling), whereas in all other Aprils, including 1998 a warm El Niño year, April upwelling was either near zero or positive. Upwelling in May 2000 was significantly less than in

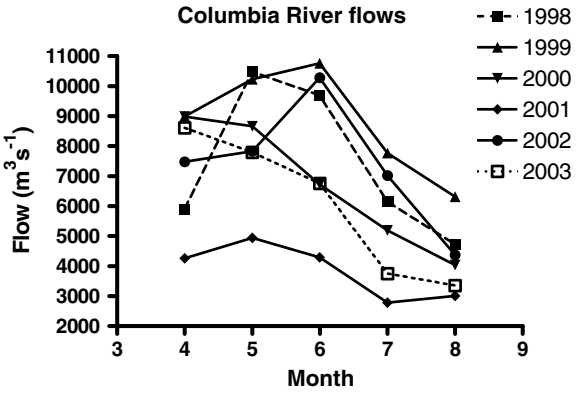


Fig. 4. Average monthly Columbia River flows from April to August 1998–2003.

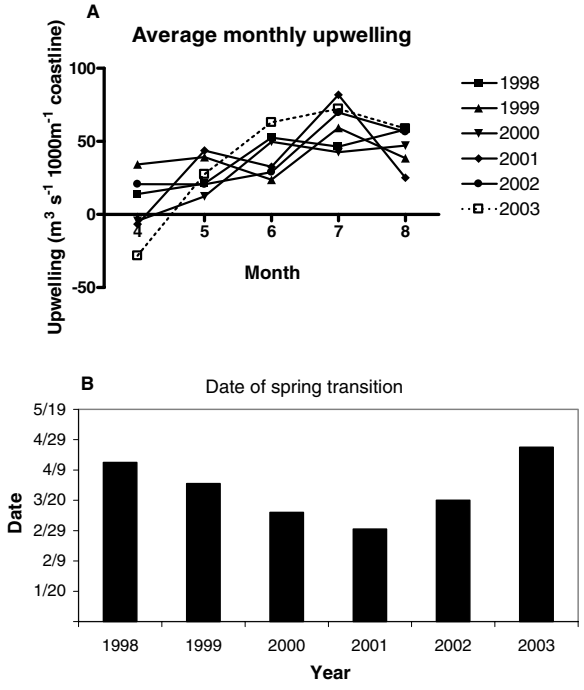


Fig. 5. (A) Average monthly upwelling from April to August at 45°N 125°W. (B) The date of the spring transition off Oregon and Washington, 1998–2003.

Table 2
Median monthly upwelling values at 45N, 125W from 1998–2003 and significant differences (Kruskal–Wallace test) among years

Month	p-Value	Year						Significant annual differences
		1998	1999	2000	2001	2002	2003	
April	0.0002	11.5	43.0	−4.5	5.0	26.5	−23.5	2003 < 1999 and 2002
May	0.0661	15.0	27.0	5.0	44.0	15.0	28.0	2000 < 2001
June	0.0033	46.5	19.5	63.0	28.5	24.5	55.0	1999 < 2003
July	0.0455	47.0	44.0	44.0	75.0	56.0	70.0	None found
August	0.0087	56.0	24.0	51.0	30.0	50.0	52.0	2001 < 1998 and 2002

Significant annual differences identified by Dunns multiple comparison test ($p < 0.05$).

2001 (Dunn's multiple range test, $p < 0.05$) (Table 2) and June 2003 had significantly higher upwelling than June 1999 (Dunn's multiple range test, $p < 0.05$). There were no interannual upwelling differences for July, but upwelling in August 2001 was lower than in either August 1998 or 2002 (Dunn's multiple range test, $p < 0.05$) (Table 2). There were also large variations in the date of the spring transition from 1998 to 2003, with 2000 and 2001 having very early transitions (early March) and 2003 having a very late transition date (April 24) (Fig. 5B).

Surface physical conditions fluctuated much more than those at depth. There were significant differences in annual sea surface (1-m) temperatures (SSTs), salinities (SSSs), and densities (SSDs) (Kruskal–Wallace test, $p < 0.0001$) during the study period (Fig. 6, Table 3), with differences in SSS related to Columbia River flow. For example, salinities were generally highest in 2001 when Columbia River flow was lowest.

Monthly SST, SSS, and SSD showed significant differences among years, with differences depending on the variable and year. For example, there were significant differences in monthly SSTs for all years (Kruskal–Wallace, $p < 0.0001$) but significant differences in among-year monthly SSSs were identified only for July (Kruskal–Wallace, $p < 0.0001$), with 2001 having higher salinities than 1999 and 2003. Monthly SSDs were significantly different among years (Kruskal–Wallace, $p < 0.0001$), and tracked surface temperatures, indicating that changes in temperature mainly determine water density differences.

Deep (50-m) water temperatures, salinities, and densities fluctuated significantly off Oregon and Washington from 1998 to 2003 (Kruskal–Wallace test, $p < 0.0001$; Fig. 7, Table 3), indicating fluctuating water masses during the study period. Much of the annual difference in temperatures at 50-m depth can be attributed to the high temperatures associated with the 1998 El Niño. However, in 2002 we observed an intrusion of cold, low-oxygen, Subarctic water that originated in the North Pacific (Fig. 7; Wheeler et al., 2003).

Monthly 50-m temperatures, salinities, and densities showed significant differences among years, but differences depended on the variable and year (Fig. 7). All monthly 50-m temperatures were significantly different (Kruskal–Wallace, $p < 0.0001$), with differences attributed to high temperatures in 1998 (all months) and 2003 (just April and May), and cold temperatures in 2002 (all months) (Dunn's multiple range test, $p < 0.05$). Significant differences in monthly 50-m salinities were only identified for April and May (Kruskal–Wallace, $p < 0.0001$). Monthly water densities were similar to salinities, with significant differences in April and May (Kruskal–Wallace, $p < 0.0001$), but not June and July. In summary, during the six-year study period, physical characteristics of 50-m water were most variable during the months of April and May.

3.2. Annual abundance of predators

Overall densities of the four predatory fishes off the Columbia River varied significantly during the six study years (Kruskal–Wallace, $p < 0.0001$) (Fig. 8). Highest average densities were observed in 1998 [$242(10^6\ m^3)^{-1}$] and lowest in 1999 [$43(10^6\ m^3)^{-1}$]. Pacific hake was the most abundant predator overall, but not for all years. In 2000 and 2001, both species of mackerel and spiny dogfish were more abundant than Pacific hake (Fig. 8).

Only two predatory species (Pacific hake and chub mackerel) showed significant among-year differences in density (Kruskal–Wallace, $p < 0.0001$) during the study period. Pacific hake showed the largest changes in annual average density, ranging from a high of 226.4 $(10^6\ m^3)^{-1}$ in 1998 to a low of 1.6 $(10^6\ m^3)^{-1}$ in 2000. In 2003, Pacific hake were again abundant with an annual average of 116.6 $(10^6\ m^3)^{-1}$. Other predators had average annual densities much lower than that of Pacific hake. While spiny dogfish had high average annual

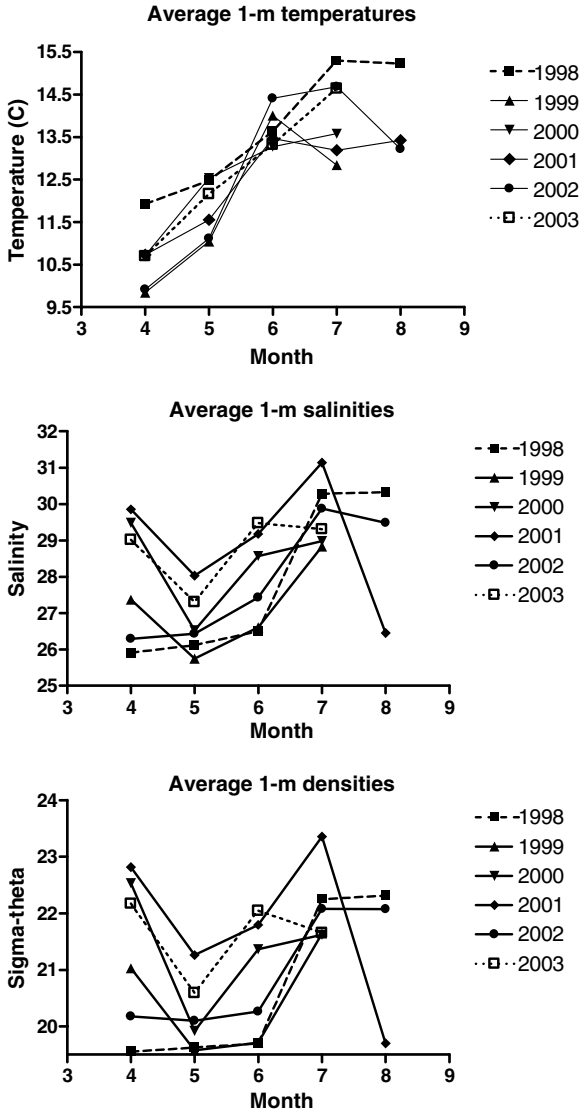


Fig. 6. Average 1-m temperatures, salinities and densities off the Columbia River, April–August, 1998–2003.

Table 3
Annual median temperatures, salinities, and densities at 1-m and 50-m depths off the Columbia River 1998–2003 and significant differences (Kruskal–Wallace test) among years

Depth (m)	Parameter	p-Value	Year						Significant annual differences
			1998	1999	2000	2001	2002	2003	
1	Temperature	<0.0001	14.01	11.84	13.27	12.91	13.57	13.1	1998 > all other years; 1999 < 2002 and 2003
1	Salinity	0.0036	29.68	28.18	29.2	30.29	29.43	29.62	1999 < 2001
1	Density (σ–θ)	0.003	21.94	21.24	21.99	22.82	22.05	22.3	1998 and 1999 < 2001
50	Temperature	<0.0001	8.73	8.05	8.33	7.81	7.42	8.22	1998 > 1999, 2001, 2002, 2003; 1999 > 2002; 2000 > 2001, 2002; 2001 > 2002; 2002 < 2003
50	Salinity	<0.0001	33.17	33.22	33.08	33.16	33.35	32.75	2003 < 1999, 2000, 2001, 2002
50	Density (σ–θ)	<0.0001	25.72	25.86	25.71	25.86	26.1	25.47	1998 < 2002; 2000 < 2002; 2003 < 1999, 2001, 2002

Significant annual differences identified by Dunn's multiple comparison test ($p < 0.05$).

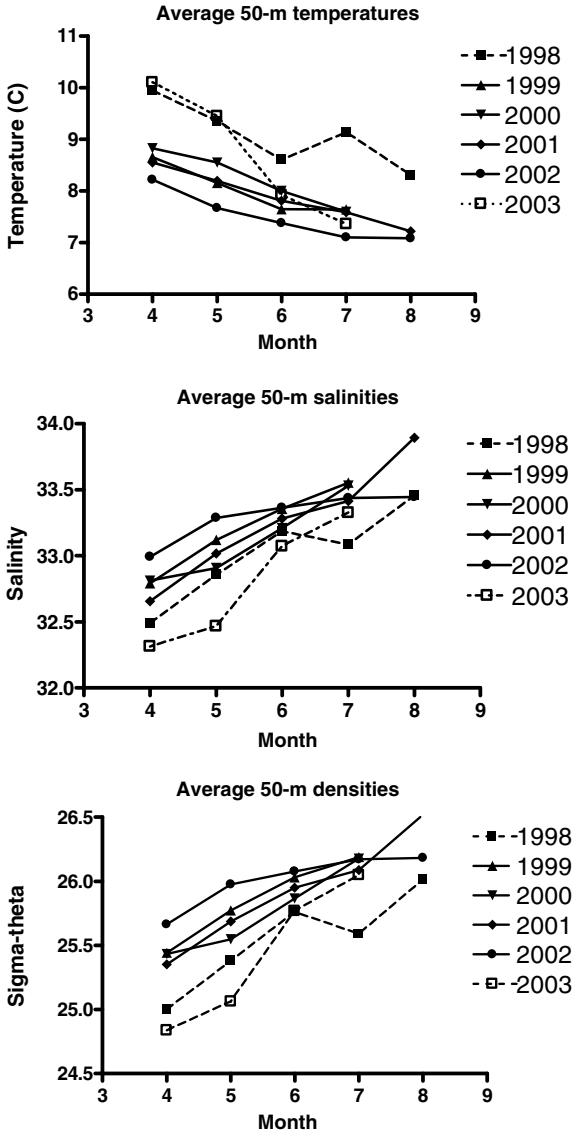


Fig. 7. Average 50-m temperatures, salinities and densities off the Columbia River, April–August, 1998–2003.

densities in 2000 and 2001 at 33.8 $(10^6\,m^3)^{-1}$ and 37.6 $(10^6\,m^3)^{-1}$, respectively, densities averaged <3 $(10^6\,m^3)^{-1}$ during other years. Annual catches of spiny dogfish were not significantly different (Kruskal–Wallace, $p = 0.319$), because catches were generally low (except for two extremely large catches). Highest average annual abundance of jack mackerel was 23 $(10^6\,m^3)^{-1}$ in 2001, but densities averaged <17 $(10^6\,m^3)^{-1}$ during all other years; annual densities were not significantly different (Kruskal–Wallace, $p = 0.456$). Chub mackerel were most abundant in 1998 [10.9 $(10^6\,m^3)^{-1}$] and least abundant in 2002 [0.10 $(10^6\,m^3)^{-1}$], with annual densities significantly different (Kruskal–Wallace, $p < 0.0001$).

Generally, when Pacific hake were abundant, other predators were not (Fig. 8). For example, lowest densities of jack mackerel and spiny dogfish occurred in 1998 and 2003, when Pacific hake were most abundant. During years when Pacific hake had relatively low densities (2000–2001), spiny dogfish and jack mackerel had their highest densities.

Predators also showed large biweekly variations in density (Fig. 9). In 1998, for example, Pacific hake had its highest densities around 28 May (our first survey), and although hake densities varied widely, they

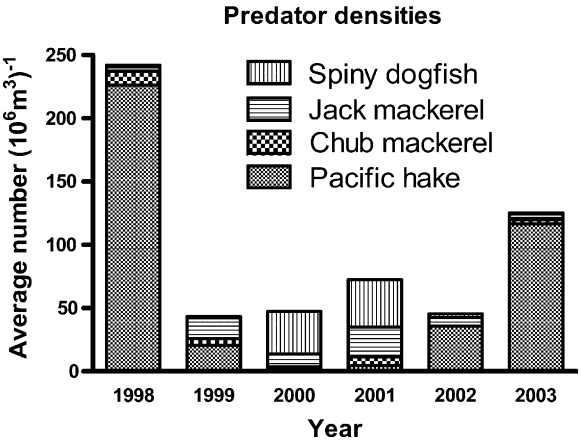


Fig. 8. Average annual densities of four pelagic predatory fishes collected at night by surface trawls off the Columbia River during spring/early summer 1998–2003.

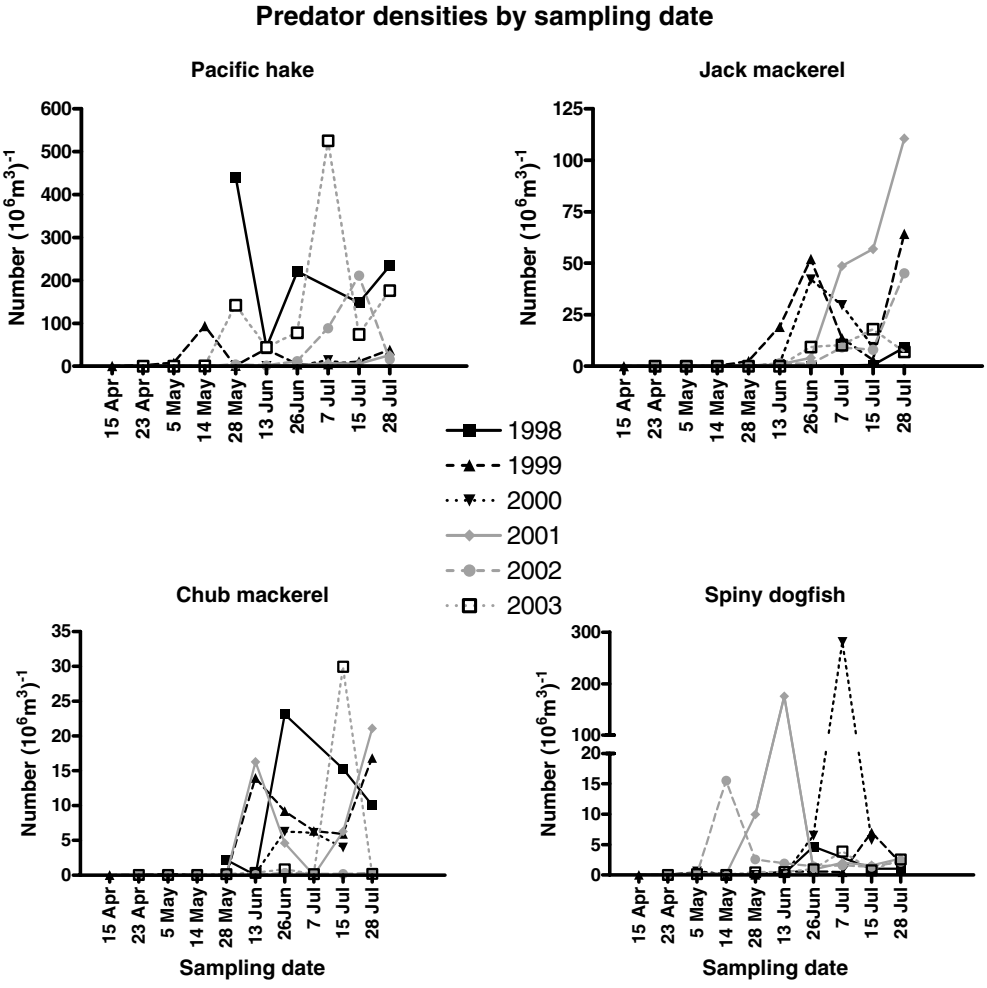


Fig. 9. Average densities of four predatory fishes collected at night by surface trawls off the Columbia River by sampling date, 1998–2003.

remained relatively high through July (Fig. 9). In 2003, Pacific hake were again abundant, and densities showed a pattern similar to 1998, becoming relatively high in late May and remaining high through July. However, unlike 1998, Pacific hake densities in 2003 were highest in early July. Both mackerel species showed a general pattern of very low numbers in April and May, increases in June, and highest densities in July (Fig. 9). Chub and jack mackerel had their highest densities of the entire study period mid-July 2003 [29.9 $(10^6\ m^3)^{-1}$] and late-July 2001 [110.6 $(10^6\ m^3)^{-1}$], respectively. Spiny dogfish densities showed three peaks during different years: early-July 2000, mid-June 2001, and mid-May 2002 (Fig. 9). However, these high densities can be attributed to single large catches of small specimens off Willapa Bay. Length/frequency analysis (not shown) revealed that most spiny dogfish catches on these cruise dates were small juveniles, indicating that Willapa Bay is probably a spiny dogfish birthing area. Ignoring the very large catches of juvenile spiny dogfish in July 2000 and June 2001, overall densities of spiny dogfish were low [$\leqslant 15\ (10^6\ m^3)^{-1}$] during all years and sampling dates.

3.3. Forage fishes

Forage fish were not abundant during 1998 or 1999, but were very abundant from 2000 to 2003 (Fig. 10). Large variations in density were seen in all four forage fish species, except Pacific sardine, but only northern anchovy increased from year-to-year from 1998 to 2003. Pacific herring and whitebait smelt densities were highest in 2001, and Pacific sardine in 2003. Densities of northern anchovy, Pacific herring, and whitebait smelt were significantly different among years (Kruskal–Wallace, $p < 0.05$), but not Pacific sardine (Kruskal–Wallace, $p = 0.0604$).

There was a strong seasonal component to forage fish densities. For example, northern anchovy and whitebait smelt densities were generally highest during late April and May, and then decreased as the summer progressed. Pacific sardine was most abundant in late June and July (Fig. 11). In 2003, both northern anchovy and whitebait smelt were abundant in May, but we caught few in June or July when, perhaps not coincidentally, densities of predatory fishes (Pacific hake and mackerel) were highest (Fig. 9). Peak densities of Pacific herring did not follow any specific trend, with time periods of peak abundance varying among years (Fig. 11). Most usually, highest densities of Pacific herring were observed in June (Fig. 11).

3.4. Juvenile salmonids

Densities of juvenile salmonids captured in the Predator Study varied widely (Fig. 12A), and all salmon showed significant density differences among years (Kruskal–Wallace, $p < 0.05$). Highest average juvenile

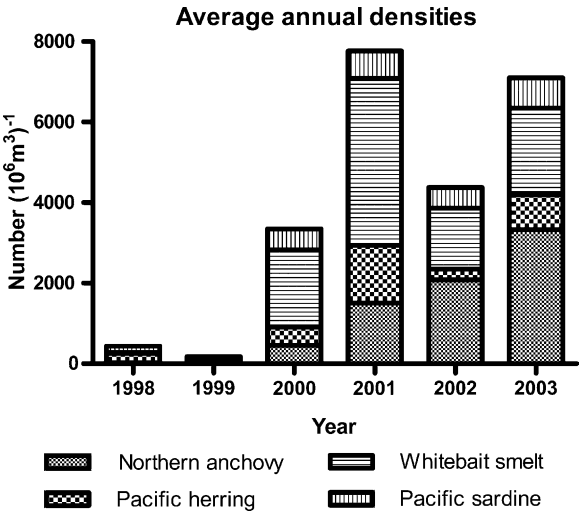


Fig. 10. Annual average densities of forage fish captured at night by surface trawls off the Columbia River, 1998–2003.

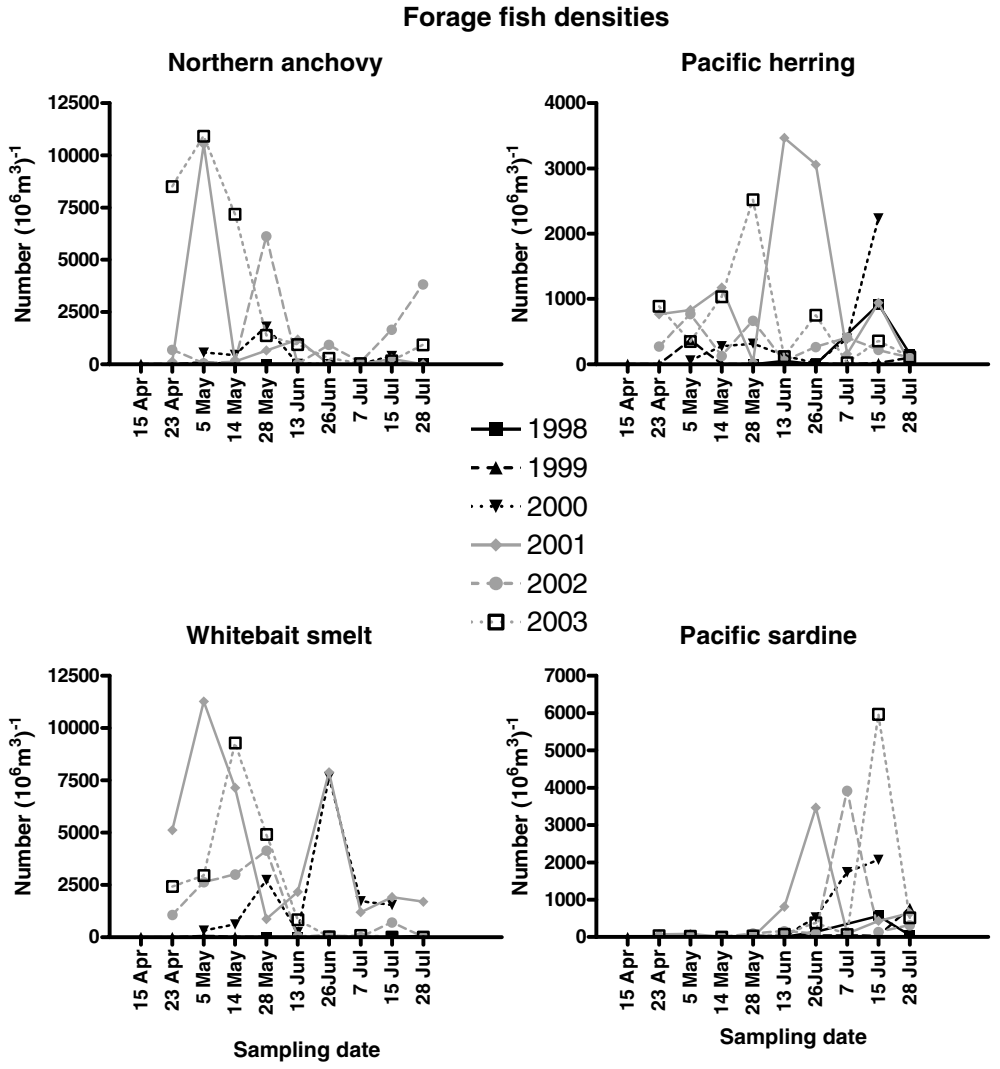


Fig. 11. Average densities of four forage fish species collected at night by surface trawls off the Columbia River by sampling date, 1998–2003.

salmon densities occurred in 2001, when many juvenile chum and 1.0-age Chinook salmon were captured. Juvenile coho salmon were found at very low annual densities except in 2000 and 2001. Sockeye salmon was rarely captured; densities of juvenile steelhead (*O. mykiss*) and cutthroat (*O. clarkii*) were so low they are not shown in Fig. 12A.

Juvenile salmon showed two seasonal abundance peaks, one in May and another in July/August, although neither peak was evident every year (Fig. 12B). Generally, the peak in May reflected the migration of yearling (1.0-age) salmon smolts and juvenile chum salmon. The July/August peak primarily reflected the migration of subyearling (0.0-age) Chinook salmon.

3.5. Spatial distribution of predatory fishes, forage fishes, and juvenile salmonids

The spatial distributions of the four predatory fishes differed by species and by year (Fig. 13). For example, Pacific hake were abundant nearshore (0–10 km) during 1998, a year with a late spring transition and with the warmest temperatures at both the sea-surface and 50-m (Figs. 6 and 7). In 1999, a relatively cold year, hake

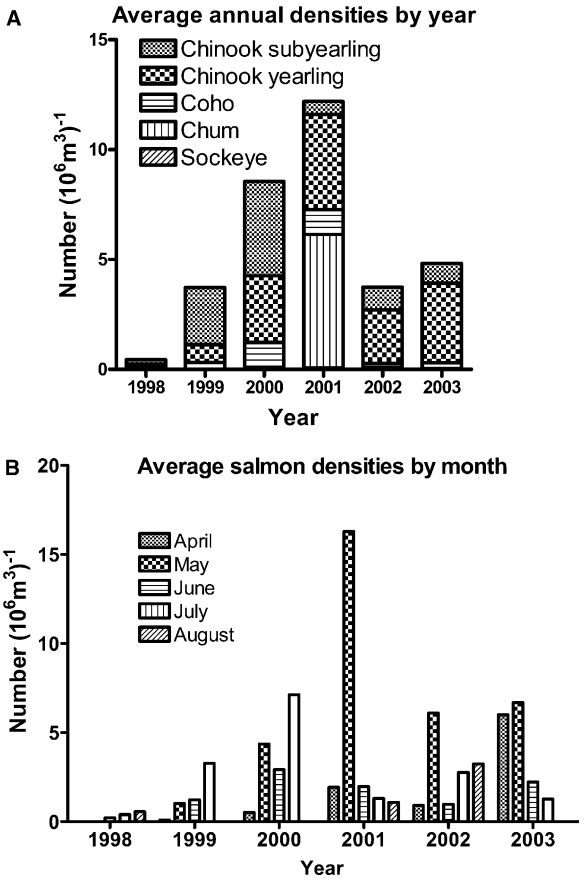


Fig. 12. (A) Average annual densities of juvenile salmonids captured at night by surface trawls off the Columbia River, 1998–2003. (B) Average densities of juvenile salmonids captured by surface trawls off the Columbia River by month, 1998–2003.

were not very abundant and were mainly caught offshore (40–50 km). In 2003, another warm year, Pacific hake were again distributed nearshore, within 10–20 km of the coast (Fig. 13).

Jack mackerel and chub mackerel had similar spatial distributions; they were most abundant at 30–40 and 20–30 km from shore, respectively; few were captured nearshore (<10 km) or far offshore (50–60 km) (Fig. 13). Spatial patterns for both species appeared to change annually, with peak abundance closer to shore during warm years and farther offshore during cold years. For example, during 1998, a warm year, chub mackerel were caught within 10 km of the coast but their highest average densities were 20–30 km offshore. In 1999, a cold year, chub mackerel were captured no closer than 20 km from the coast, and peak densities were 30–40 km offshore.

Spiny dogfish distribution was primarily nearshore-oriented, with first and second highest densities found less than 10 and 10–20 km from shore, respectively (Fig. 13). High nearshore densities in 2000 and 2001 were strongly skewed by single large catches of small juveniles just off Willapa Bay, but their overall distribution was still nearshore; very few spiny dogfish were captured between 40 and 60 km from shore.

Catches of most forage fishes showed significant inshore/offshore trends. All but sardine had highest densities nearshore and very low densities offshore (Fig. 14); no forage fishes were caught at the station farthest offshore (50–60 km), which was beyond the continental shelf (Fig. 14). Northern anchovy showed a high affinity for the very nearshore environment after accounting for year (extra sum of squares F test, $F = 11.43$, $p < 0.001$); highest densities during all years were <10 or 10–20 km from shore (Fig. 14). Northern anchovy densities dropped quickly beyond 20 km from the coast, with very few captured 40–50 km from shore. Pacific herring also showed a preference for nearshore water, with highest densities 0–30 km offshore and very low densities farther offshore, but this relationship was not significant after accounting for year

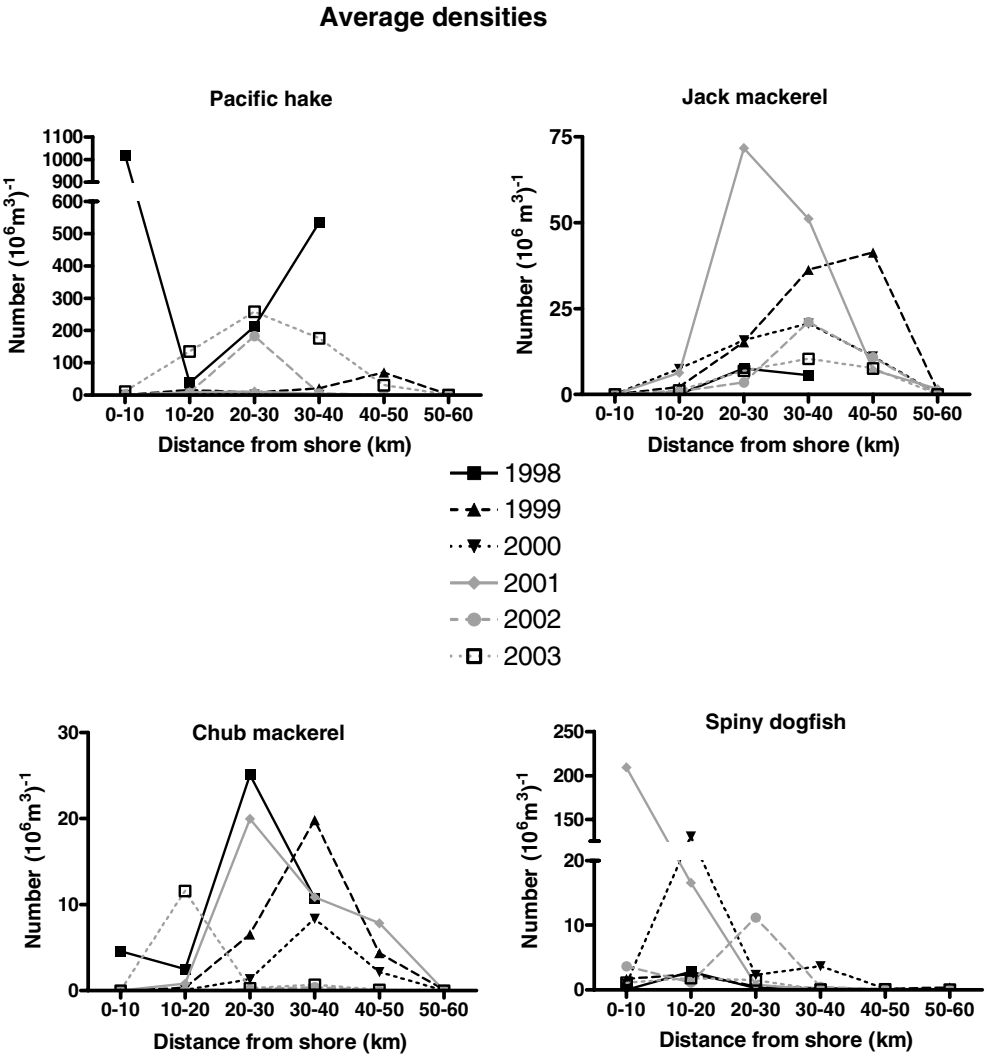


Fig. 13. Average annual densities of four predatory fishes collected at night by surface trawls off the Columbia River at different distances from shore, 1998–2003.

($F = 1.638$, $p = 0.1737$) (Fig. 14). While highest herring density was observed within 10 km of the coast in 2003, highest densities during other years were 20–30 km (2001) or 10–20 km from the coast (2000). Very few Pacific herring were captured over the slope, 40–60 km off the coast. Whitebait smelt densities were significantly related to distance from shore ($F = 6.45$, $p = 0.0169$), with highest densities 10–20 km from the coast during all years (Fig. 14); few were captured beyond 30 km and none 40–60 km from shore. Pacific sardine were distributed differently from the other forage fishes. There was no consistent relationship between sardine densities and distance from shore after accounting for year ($F = 0.1512$, $p = 0.9872$). For example, in 2000 most sardine were captured within 20 km of the coast, but in 2001 highest densities were 40–50 km from the coast (Fig. 14).

Juvenile salmon captured during the Predator Study (nighttime catches) showed distribution patterns that varied widely depending on species and year (Fig. 15). In general, most juvenile salmon species/and age-classes had high densities nearshore, but no salmon species/age-class had densities that were significantly related to distance from shore after accounting for year (extra sum of squares F test, $p > 0.05$). Highest densities of juvenile coho salmon in 2000 were 40–50 km from shore. Both 0.0-age Chinook and chum salmon showed weak negative relationships between distance offshore and density. This was observed for densities of 0.0-age

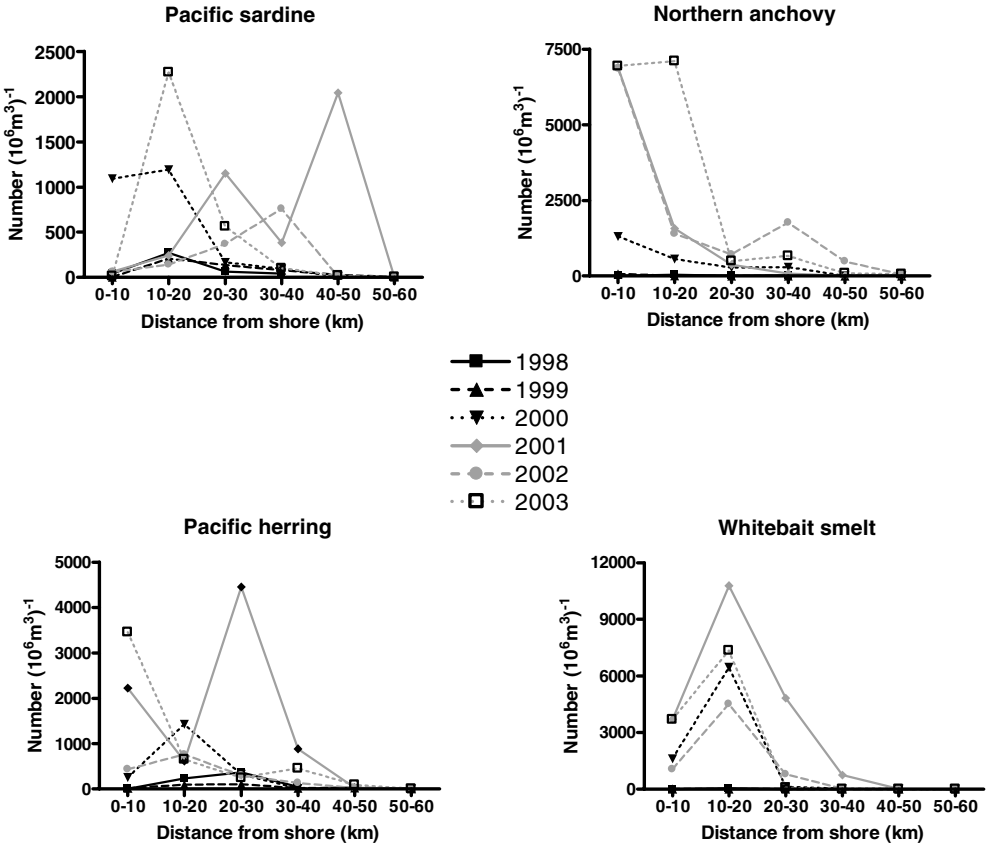


Fig. 14. Average annual densities of four forage fishes collected at night by surface trawls off the Columbia River at different distances from shore, 1998–2003.

Chinook salmon for 1999 and 2000, but not for other years. Nearly all juvenile chum salmon captured were taken in 2001, so it was unclear whether the observed nearshore distribution pattern re-occurred annually. Finally, 1.0-age Chinook salmon densities showed no clear relation to distance offshore, with similar densities both offshore and nearshore during most years. Most of the juvenile salmon captured during the Predator Study were undoubtedly recent migrants from the Columbia River. As such, their marine distribution was influenced by the shape, direction, and strength of the Columbia River plume.

June data from the Oregon/Washington coast-wide BPA Plume Study (daytime catches) provide juvenile salmonid coastal distributions away from the Columbia River. These data show onshore/offshore distribution patterns similar to those from the Predator Study for 0.0-age Chinook salmon, but not for other salmonids species/age groups. For example, 0.0-age Chinook salmon densities were highest nearshore (1–20 km), and this relationship was significant after accounting for year ($F = 8.52$, $p = 0.0067$) (Fig. 16). Unlike the Predator Study catches, chum salmon densities from the Plume Study were highest 40–50 km offshore (Fig. 16) but this relationship was not significant ($F = 2.464$, $p = 0.1273$). Yearling (1.0-age) Chinook salmon in the Plume Study showed a significant relationship between density and distance from shore ($F = 4.83$, $p = 0.0364$), with densities highest between 10–30 km and gradually decreasing to zero at 60 km offshore. Finally, coho salmon in the Plume Study were most abundant from 10 to 50 km offshore, and not related to distance from shore ($F = 2.68$, $p = 0.1123$).

3.6. Predator densities, oceanography, forage fishes, and juvenile salmonids

Annual average Pacific hake densities were highest closest to shore (Fig. 9), when the spring transition was late (Fig. 5A) and spring (April–May) temperatures were warmest at the 50-m depth. These conditions

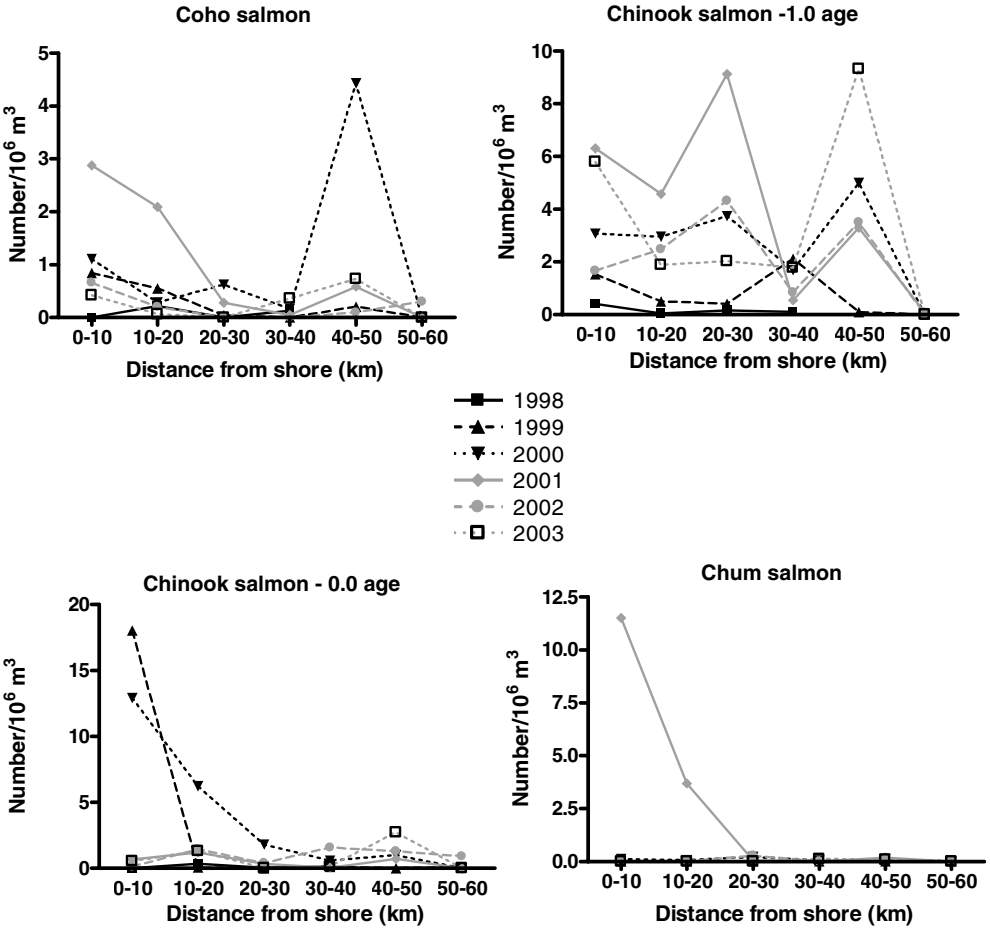


Fig. 15. Average annual densities of juvenile salmonids collected at night by surface trawls off the Columbia River at different distances from shore, 1998–2003.

occurred in 1998 and 2003 (Fig. 7A). As a result, annual average Pacific hake densities were strongly related to the spring transition (GLM, $p = 0.028$, adjusted $R^2 = 67.36\%$) (Fig. 17). High hake catches in May/June occurred in 1998 and 2003, years with a late spring transition.

Annual jack mackerel densities were highest when the spring transition was early (Fig. 17), but this relationship was not statistically significant (GLM, $p = 0.0729$, $R^2 = 49.2\%$). Jack mackerel densities varied inversely to Pacific hake densities, and were highest during cool ocean years (1999–2002). Annual chub mackerel densities were not related to the spring transition or any other physical or biological variables (GLM, $p > 0.10$). Spiny dogfish densities were generally very low and showed no relationship with oceanographic conditions.

We found no statistical relationship between predatory fish abundance and forage fish abundance (GLM, $p > 0.05$). However, seasonal abundance of predatory fishes appears to be inversely related to forage fish abundance. For example, both whitebait smelt and northern anchovy were generally most abundant in early spring (May), while predatory fishes usually peaked later (June and July) (Fig. 9). After the arrival of predatory fishes, particularly Pacific hake in late May 2003, abundance of northern anchovy and whitebait smelt declined dramatically. This was most evident for whitebait smelt, which were abundant in May 2003 but virtually disappeared from our catches by late June–July 2003. Examination of Pacific hake stomach contents indicated they were feeding on whitebait smelt and anchovy in 2003.

Annual densities of Pacific hake (Fig. 8) generally trended oppositely to those of juvenile salmonids (Fig. 12), but it was unclear whether Pacific hake abundance directly affected juvenile salmonid abundance.

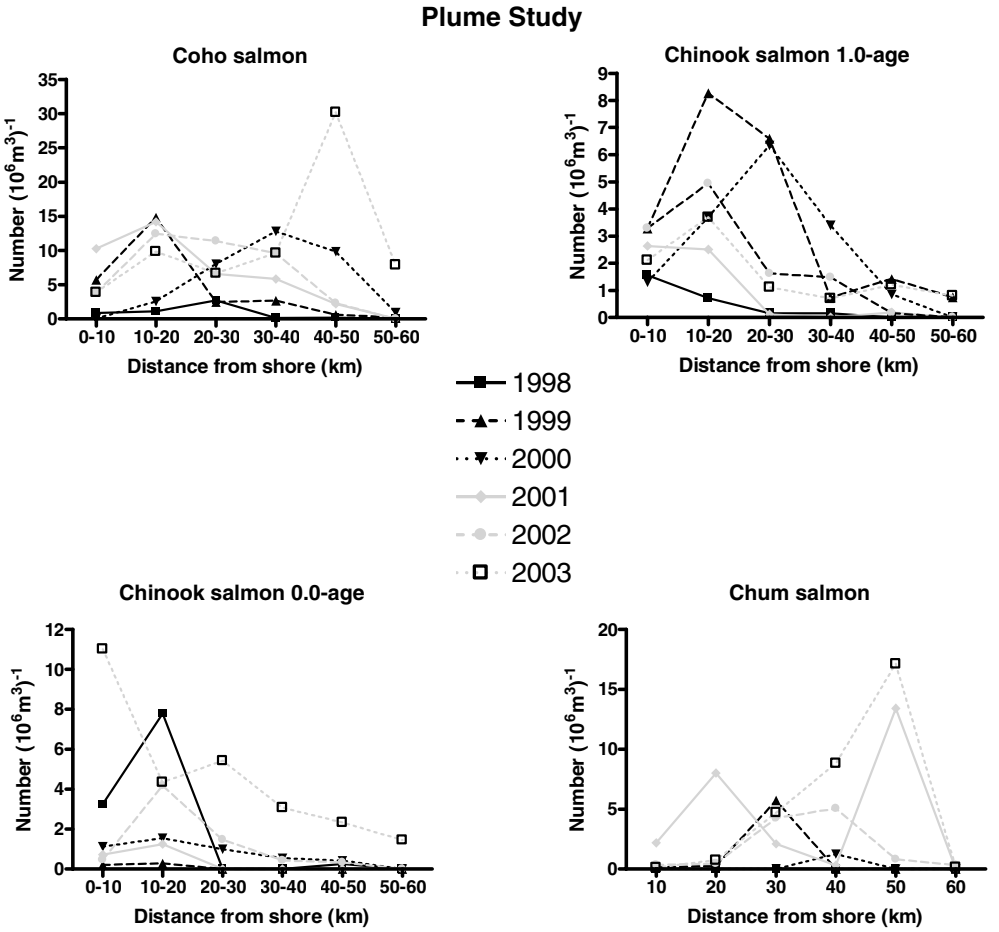


Fig. 16. Average annual densities of juvenile salmonids collected during BPA Plume study (daylight) by surface trawls in June off the Oregon/Washington coast at different distances from shore, 1998–2003.

Statistical analysis indicated a possibly negative relationship between juvenile salmonid densities and Pacific hake densities (GLM, $p = 0.064$, $R^2 = 52.14\%$) but this was not significant. However, there was a significant negative relationship between annual average May/June densities of hake and OPI coho salmon percent survival (GLM, $p = 0.030$, $R^2 = 57.01\%$) (Fig. 18).

Analysis of presence/absence of juvenile salmonids versus predator and forage fishes sampled by trawl indicated strong interactions among species (Table 4). With the exception of spiny dogfish, when predators occurred in a trawl, salmonids usually did not (χ^2, $p < 0.05$). Conversely, there was a positive interaction (i.e., species co-occurred more than expected) between the occurrence of juvenile salmonids and northern anchovy, Pacific herring, and whitebait smelt (χ^2, $p < 0.05$). There was no relationship between the occurrence of juvenile salmonids and Pacific sardine in the catches (χ^2, $p = 0.396$).

4. Discussion

This is the first localized, high-frequency sampling study of pelagic fishes off the coast of Oregon and Washington. It is also the only study that has collected pelagic fishes at night, thus eliminating the bias of daytime surface trawling (Krutzikowsky & Emmett, 2005). This and other studies (Brodeur & Pearcy, 1986; Emmett & Brodeur, 2000) show that the abundance and distribution of large, pelagic, predatory fishes off Oregon and Washington fluctuates widely, both annually and seasonally. These changes in predatory fish abundance and distribution appear to be strongly linked to the date of the spring transition, as well as to changes in ocean

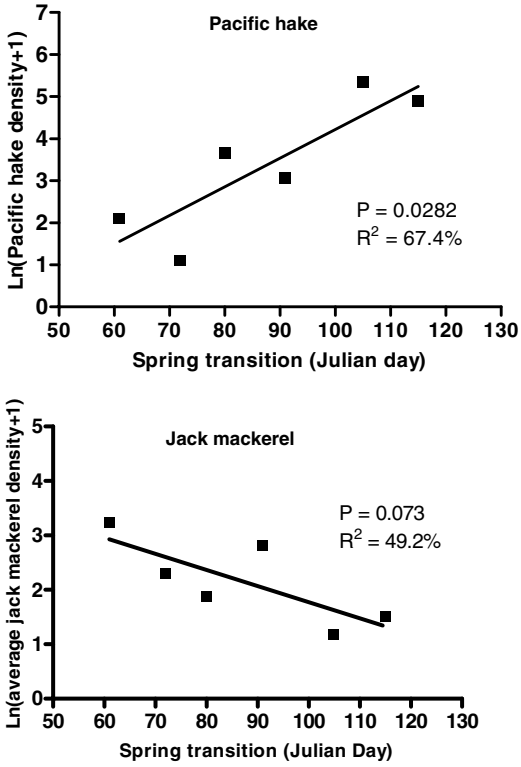


Fig. 17. Relationship between annual Pacific hake (*Merluccius productus*) and jack mackerel (*Trachurus symmetricus*) densities identified by surface trawling at night off the Columbia River, 1998–2003, and date of the spring transition.

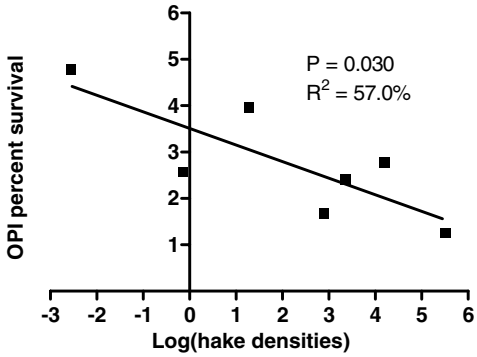


Fig. 18. Relationship between average May/June densities of Pacific hake (*Merluccius productus*) caught off the mouth of the Columbia River, 1998–2004, and coho salmon (*Oncorhynchus kisutch*) OPI percent marine survival.

temperature (Brodeur & Pearcy, 1986; Dorn, 1995; Ware & McFarlane, 1995) and prey abundance (Benson et al., 2002). Warmer ocean temperatures (at depth and the surface) and the date of the spring transition influenced the movement of Pacific hake and probably jack mackerel in Northwest waters.

Our catches indicated Pacific hake were distributed similarly to the coastal distribution of hake identified by NMFS triennial acoustic surveys (Helser, Methot, & Fleischer, 2004). These surveys found hake to be relatively abundant in the Columbia River region in 1998 and 2003, but not in 2001 (Helser et al., 2004). Our low hake catches in 2001 appears to have been related to both reduced Pacific hake stock size, and slower hake migration to the north because of cooler ocean temperatures and slower northerly currents.

Table 4
Percent surface trawls with and without juvenile salmonids and other fishes along with statistical differences (χ^2 test, with Yates correction) between the occurrence of juvenile salmonids and other fishes in trawls

	Pacific hake	Chub mackerel	Jack mackerel	Spiny dogfish	Northern anchovy	Pacific herring	Whitebait smelt	Pacific sardine
With juvenile salmonids	11.87%	4.68%	7.69%	10.87%	20.74%	31.61%	21.57%	17.22%
Without juvenile salmonids	22.07%	11.70%	16.05%	10.37%	19.57%	27.59%	13.71%	23.75%
χ^2	10.3	11.02	10.1	2.74	7.86	28.06	36.35	0.72
p-Value	0.0013	0.0009	0.0015	0.0980	0.0051	0.0000	0.0000	0.3960
Relationship	Negative	Negative	Negative	No	Positive	Positive	Positive	No

Relationship variable indicates whether predator or forage fish species occurrences were significantly negative (i.e., <expected) or positive (i.e., >expected), or there was no relationship.

The Pacific hake population in 2003 was estimated to be between 2.7 and 4.2 million tons (Helser et al., 2004). With such a large biomass, hake movement and feeding habits undoubtedly affect other biological resources in the California Current ecosystem (Jay, 1996), including Pacific herring off British Columbia, Canada (Ware & McFarlane, 1995), pink shrimp off Oregon (Hannah, 1995), and probably anchovy and whitebait smelt off Oregon and Washington (Predator Study).

One reason we conducted our sampling at night was that we suspected interactions among predators (particularly hake), forage fish, and juvenile salmonids would be most evident during nocturnal hours. Juvenile salmonids off Oregon appear not to migrate vertically, but remain near the surface (Emmett et al., 2004; Pearcy & Fisher, 1988). They are thus more likely to interact with diel migrating predatory and forage fishes at night. Preliminary stomach analyses indicated that hake, mackerel, and spiny dogfish, feed actively at night, with euphausiids and forage fishes the primary prey. Alton and Nelson (1970) concluded that Pacific hake feed primarily between sunset and sunrise, but Livingston (1983) found feeding during both day and night. Rexstad and Pikitch (1986) found evidence that hake eat more at night, but their data may have been influenced by an El Niño (possibly altering the food web). We believe that hake eat during both day and night, migrating vertically to follow prey (euphausiids and forage fish) and feeding at night if prey is abundant.

Ongoing stomach analyses have identified few juvenile salmonids in any predator stomachs. As noted by Holtby, Andersen, and Kadawaki (1990) in British Columbia waters, the scarcity of salmonids relative to forage fish would make it unlikely that salmon would constitute a significant fraction of Pacific hake diet. However, even if juvenile salmon are incidental prey, hake biomass is sufficiently large to cause substantial juvenile salmonid mortality. The significantly negative relationship between hake densities and coho salmon marine survival supports the hypothesis that predation when salmon first enter the ocean is a major source of their marine mortality.

The presence of Pacific hake and other predators in a trawl was negatively related to the presence of salmon smolts and forage fishes, indicating that predators could be altering the distributions of these fishes. Large piscivorous fishes feed most effectively in clear, non-turbid water, while planktivorous fish feeding is relatively unaffected by turbidity except at very high levels (De Robertis, Ryer, Veloza, & Brodeur, 2003). So, we suspect that ocean turbidity may play a critical role in the onshore/offshore distributions and interactions among predators, salmonids, and forage fish. Predatory fishes, such as Pacific hake, are generally found farther offshore (Fig. 13) in low turbidity (high transmissivity) waters (Fig. 19), while most salmonid and forage fish species are found primarily inshore (Figs. 14–16) in higher turbidity (low transmissivity) waters. In 2000 and 2001, when hake densities were very low, both Pacific herring and whitebait smelt were distributed farther from shore, perhaps because these areas offered less predation pressure and more turbid ocean conditions.

Our intensive field sampling was designed to identify when pelagic predators and forage fishes first become abundant off the Columbia River. Our data indicate that hake move into nearshore waters during years with warm springs (weak upwelling) and clear (low primary production and low turbidity) waters (e.g., 1998 and 2003) (Emmett, pers. obs). Methot and Dorn (1995) noted that hake usually move onto the shelf in June. Our data indicate that in years when the ocean is warm in spring, especially at depth, hake will move onto the Oregon/Washington shelf by late May, which is also the peak migrational period of yearling salmonids leaving the Northwest coastal rivers. May is also when forage fishes peaked in abundance. We hypothesize that the spring migration of juvenile salmon from the Columbia River is a life history strategy to minimize marine predation.

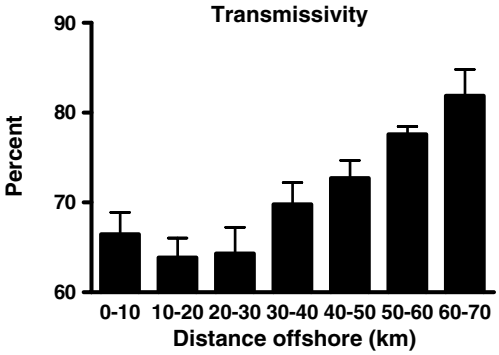


Fig. 19. Average surface (1-m depth) percent transmissivity (water clarity) observed at BPA Plume Study surface trawling stations off Oregon and Washington by distance from shore, 2002–2003.

Migrating juvenile salmon can minimize marine predation by arriving in the ocean before predators become abundant and when alternative prey (forage fishes) are abundant.

There was evidently a shift in oceanographic conditions off Oregon and Washington in 1999. This shift caused changes in the zooplankton species composition and abundance (Peterson & Schwing, 2003) and in marine survival of salmon (Williams et al., 2005). Our data indicate that the composition and abundance of predatory fish species changed in 1999, and that of forage fish changed by 2000. The distributions and densities of Pacific hake and jack mackerel were much different after 1998. However, in 2003, ocean conditions were similar to those of 1998: the spring transition was late, spring ocean temperatures were warm, and Pacific hake were again relatively abundant on the shelf. While predatory fish densities and distributions were similar in 1998 and 2003, forage fish densities were not. We believe this was because predatory fishes are long-lived and migrate to the study area (except dogfish) during the spring and summer, while forage fishes (except sardine) are primarily local spawners. As such, predators can immediately shift their population distributions with changing ocean conditions, while the short-lived forage fishes must respond by changing recruitment patterns. Forage fish population abundances showed a 1-year lag between changed oceanographic condition/predator densities (1999) and increasing densities (2000). Because of their different life history strategies, forage fish appear to respond differently than predatory fishes to the fluctuating oceanographic climatic conditions.

Northwest coastal ocean temperatures, primary productivity, and ocean currents are linked, and all affect the distributions and abundances of predatory fish species in the Columbia River plume. Warmer ocean temperatures at 50-m in spring indicate a late spring transition and either continued northerly transport (Davidson current) or slow southerly transport. As Dorn (1995) showed, during years with a warm spring Pacific hake are transported more quickly to the north because northerly currents are stronger: conversely, during cooler spring years, transport is slower due to strong southerly currents. Coastal current meter data (Kosro, 2003) and altimetry data (Strub & James, 2003) collected during most of our study period support this hypothesis. During 2000 and 2002, spring coastal currents off Oregon were anomalously equatorward, and hake densities were low; in 1998 and 2003 currents were more northward (M. Kosro, Oregon State University, pers. comm.), and hake densities on the shelf were high.

Using purse seines during daylight hours, Brodeur and Pearcy (1986) observed wide fluctuations in the abundance of pelagic nekton species off Oregon and Washington from 1979 to 1984, concurrent with relatively small changes in ocean temperatures. They found chub mackerel was the dominant predatory fish during the warm ocean conditions of 1983–1984, not Pacific hake as we observed in 1998 and 2003. However, diel-migrating fishes such as hake would be underrepresented in their daylight catches. We suspect that lower densities of chub mackerel were observed in 1998 and 2003 because the west coast chub mackerel population is much lower now than in the 1980s (California Dept. Fish Game, 2002). Nevertheless, similarly to our study, Brodeur and Pearcy (1986) also observed higher predatory fish catches (mackerel and hake) and smaller forage nekton catches during warmer ocean conditions.

Rebounding Northwest salmonid runs appear to have begun with the high marine survival of juvenile salmonids that entered the ocean in 1999 (coho salmon OPI index; Williams et al., 2005). That year, overall predatory

fish abundance off Oregon and Washington, particularly Pacific hake during May and June, were considerably lower than in 1998, suggesting that reduced predatory fish abundance may be related to increased salmonid survival. Holtby et al. (1990) concluded that growth of age-1 coho salmon and fluctuations in predation intensity (number of predators) accounted for the annual variations in coho salmon marine survival on the west side of Vancouver Island, Canada. Fisher and Pearcy (1988) and Pearcy (1992) observed that low marine survival of Oregon coho salmon was not due to food shortages or slow growth in the ocean, but hypothesized that years with poor marine survival of salmon related to increased marine mortality caused by predators switching to salmon when their preferred prey were scarce. Our data support this hypothesis; when predator numbers decreased, marine survival of salmon increased. We also observed a coincidently large increase in forage fish or "alternative" prey abundance since 2000. If alternative prey are important (Fisher & Pearcy, 1988; Pearcy, 1992), then marine survival of out-migrant salmon smolts in 2003 (a warm year with many predators) will decline, but not to the extremely low levels seen during most of the 1990s, because, while predators were numerous, alternative prey (forage fish) remained abundant. Future adult salmon returns should verify whether this hypothesis is true.

In conclusion, the effects of climate variability on the ocean fisheries resources are considerable (Anderson & Piatt, 1999; Beamish, 1995; Beamish et al., 1999; Benson et al., 2002; Cushing, 1982; Field & Francis, 2002; Hare, Mantua, & Francis, 1999; Hollowed, Hare, & Wooster, 2001; Mantua, Hare, Zhang, Wallace, & Francis, 1997; McFarlane, King, & Beamish, 2000; McGowan, Cayan, & Dorman, 1998; Ware, 1995; Ware & McFarlane, 1995; Zhang, Lee, Kim, & Oh, 2000). Our research indicates that large-scale oceanographic forcing affects the movements and abundance of predatory fishes around the Columbia River plume. These predatory fishes appear to affect in turn the abundance and distribution (inshore/offshore) of forage fishes. While it is still uncertain how these fish predator/fish prey interactions affect marine survival of juvenile salmonids, circumstantial evidence suggests that these interactions are an important factor. Non-salmonid predatory and forage fishes numerically dominate the pelagic nearshore fish community in the northern California Current. Only long-term observational studies can provide the data needed to assess how species interactions affect predatory and forage fish abundances and the role these fishes play in marine survival of salmonids. This knowledge will be critically important if we are effectively and safely to manage Northwest coastal fisheries, including salmonids, during oceanic climate shifts.

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References

Alton, M. S., & Nelson, M. O. (1970). Food of Pacific hake, *Merluccius productus*, in Washington and northern Oregon coastal waters. *U.S. Fish Wildlife and Service Circular, 332*, 35–42.

Anderson, P. J., & Piatt, J. F. (1999). Community reorganization in the Gulf of Alaska following ocean climate regime shift. *Marine Ecology Progress Series, 189*, 117–123.

Arcos, D. F., Cubillos, L. A., & Núñez, S. P. (2001). The jack mackerel fishery and El Niño 1997–1998 effects off Chile. *Progress Oceanography, 49*, 597–617.

Bailey, K. M., Francis, R. C., & Stevens, P. R. (1982). The life history and fishery of Pacific whiting, *Merluccius productus*. *California Cooperative Fisheries Investigation Reports, 23*, 81–98.

Bax, N. J. (1998). The significance and prediction of predation in marine fisheries. *ICES Journal of Marine Science, 55*, 997–1030.

Beacham, T. D. (1991). *The marine survival of salmon program. Program outline and investigators summaries for 1989/90*. Pacific Biological Station, Nanaimo, B.C. Canada: Department of Fisheries and Oceans, 70p.

Beamish, R. J. (Ed.). (1995). *Climate change and northern fish populations*. *Canadian special publication of fisheries and aquatic sciences* (vol. 121). Ottawa: National Research Council of Canada.

Beamish, R. J., Noakes, D. J., McFarlane, G. A., Klyashtorin, L., Ivanov, V. V., & Kurashov, V. (1999). The regime concept and natural trends in the production of Pacific salmon. *Canadian Journal of Fisheries and Aquatic Science, 56*, 516–526.

Benson, A. J., McFarlane, G. A., Allen, S. E., & Dower, J. F. (2002). Changes in Pacific hake (*Merluccius productus*) migration patterns and juvenile growth related to the 1989 regime shift. *Canadian Journal of Fisheries and Aquatic Science, 59*, 1969–1979.

Blaxter, J. H. S., & Holliday, F. G. T. (1963). The behavior and physiology of herring and other clupeids. *Advances in Marine Biology, 1*, 261–393.

Brodeur, R. D., Fisher, J. P., Emmett, R. L., Morgan, C. A., & Casillas, E. (2005). Species composition and community structure of pelagic nekton off Oregon and Washington under variable oceanographic conditions. *Marine Ecology Progress Series, 298*, 41–57.

Brodeur, R. D., & Pearcy, W. G. (1986). Distribution and relative abundance of pelagic non-salmonid nekton off Oregon and Washington, 1979–1984. *NOAA Technical Report, NMFS No. 46*.

Borgstad, B., & Mehl, S. (1997). Interactions between Atlantic cod (*Gadus morhua*) and its prey species in the Barents Sea. In *Forage Fishes in Marine Ecosystems, Proceedings International Symposium on the Role of Forage Fishes in Marine Ecosystems* (pp. 591–615). Fairbanks, AK: University Alaska Sea Grant College Program Report No. 97-01.

California Department of Fish and Game., 2002. Review of some California fisheries for 2001: market squid, sea urchin, Dungeness crab, lobster, prawn, abalone, groundfish, swordfish and shark, coastal pelagic finfish, ocean salmon, nearshore live-fish, Pacific herring, white seabass, and kelp. *California Cooperative Fisheries Investigation Reports* 43, 13–30.

Carscadden, J. E., Frank, K. T., & Leggett, W. C. (2001). Ecosystem changes and the effects on capelin (*Mallotus villosus*), a major forage species. *Canadian Journal of Fisheries and Aquatic Science, 58*, 73–85.

Compagno, L. J. V. (1984). Sharks of the world. *FAO Fisheries Synopsis, Part 1, 125*(4).

Cushing, D. H. (1982). *Climate and fisheries*. London: Academic Press.

De Robertis, A., Ryer, C. H., Veloza, A., & Brodeur, R. D. (2003). Differential effects of turbidity on prey consumption of piscivorous and planktivorous fish. *Canadian Journal of Fisheries and Aquatic Science, 60*, 1517–1526.

Dorn, M. W. (1995). The effects of age composition and oceanographic conditions on the annual migration of Pacific whiting *Merluccius productus*. *California Cooperative Oceanic Fisheries Investigation Reports, 36*, 97–105.

Dotson, R. C., & Griffith, D. A. (1996). A high-speed midwater rope trawl for collecting coastal pelagic fishes. *California Cooperative Oceanic Fisheries Investigation Reports, 37*, 134–139.

Emmett, R. L., & Brodeur, R. D. (2000). Recent changes in the pelagic nekton community off Oregon and Washington in relation to some physical oceanographic conditions. *North Pacific Anadromous Fisheries Commission Bulletin, 2*, 11–20.

Emmett, R. L., Brodeur, R. D., Miller, T. W., Pool, S. S., Bentley, P. B., et al. (in press). Pacific sardine (*Sardinops sagax*) abundance, distribution, and ecological relationships in the Pacific Northwest. *California Cooperative Fisheries Investigation Reports*.

Emmett, R. L., Brodeur, R. D., & Orton, P. M. (2004). The vertical distribution of juvenile salmon and associated fishes in the Columbia River plume. *Fisheries Oceanography, 13*, 392–402.

Field, J. C., & Francis, R. C. (2002). Cooperating with the environment: Case studies of climate and fisheries in the northern California Current. In *Fisheries in a Changing Climate* (245–260). American Fisheries Society Symposium 32.

Fisher, J. P., & Pearcy, W. G. (1988). Growth of juvenile coho salmon (*Oncorhynchus kisutch*) off Oregon and Washington, USA in years of differing coastal upwelling. *Canadian Journal of Fisheries and Aquatic Science, 45*, 1036–1044.

Gjøsaeter, H. (1995). Pelagic fish and the ecological impact of the modern fishing industry in the Barents Sea. *Arctic, 48*, 267–278.

Grantham, B. A., Chan, F., Nielsen, K. J., Fox, D. S., Barth, J. A., Huyer, A., et al. (2004). Upwelling-driven nearshore hypoxia signals ecosystem and oceanographic changes in the northeast Pacific. *Nature, London, 429*, 749–754.

Hare, S. R., Mantua, N. J., & Francis, R. C. (1999). Inverse production regimes: Alaska and West Coast Pacific salmon. *Fisheries, 24*(1), 6–15.

Hart, J. L. (1973). Pacific fishes of Canada. *Fisheries Research Board of Canada, Bulletin*, 180.

Hannah, R. W. (1995). Variation in geographic stock area, catchability, and natural mortality of ocean shrimp (*Pandalus jordani*): some new evidence for a trophic interaction with Pacific hake (*Merluccius productus*). *Canadian Journal of Fisheries and Aquatic Science, 52*, 1018–1029.

Helser, T. E., Methot, R. D., & Fleischer, G. W. (2004). *Stock assessment of Pacific whiting in U.S. and Canadian waters in 2003*. Portland, OR: North Pacific Fisheries Management Council.

Hickey, B. M. (1989). Patterns and processes of circulation over the coastal shelf off Washington. In M. R. Landry & B. M. Hickey (Eds.), *Coastal oceanography of Washington and Oregon* (pp. 41–115). Amsterdam: Elsevier.

Hickey, B. M., & Banas, N. S. (2003). Oceanography of the U.S. Pacific Northwest coastal ocean and estuaries with application to coastal ecology. *Estuaries, 26*, 1010–1031.

Hollowed, A. B. (1992). Spatial and temporal distributions of Pacific hake, *Merluccius productus*, larvae and estimates of survival during early life stages. *California Cooperative Fisheries Investigation Reports, 33*, 100–123.

Hollowed, A. B., Ianelli, J. N., & Livingston, P. A. (2000). Including predation mortality in stock assessments: a case study for the Gulf of Alaska walleye pollock. *ICES Journal of Marine Science, 57*, 279–293.

Hollowed, A. B., Hare, S. R., & Wooster, W. S. (2001). Pacific Basin climate variability and patterns of Northeast Pacific marine fish production. *Progress in Oceanography, 49*, 257–282.

Holtby, L. B., Andersen, B. C., & Kadawaki, R. K. (1990). Importance of smolt size and early ocean growth to interannual variability in marine survival of coho salmon (*Oncorhynchus kisutch*). *Canadian Journal of Fisheries and Aquatic Science, 47*, 2181–2194.

Jay, C. V. (1996). Distribution of bottom-trawl fish assemblages over the continental shelf and upper slope of the U.S. West Coast, 1977–1992. *Canadian Journal of Fisheries and Aquatic Science, 53*, 1203–1225.

Koo, T. S. Y. (1962). Age designation in salmon. In T. S. Y. Koo (Ed.), *Studies of Alaska red salmon* (pp. 41–48). Seattle: University Wash Press.

Kosro, P. M. (2003). Enhanced southward flow over the Oregon shelf in 2002: a conduit for subarctic water. *Geophysical Research Letters, 30*(15), 8023, 1–4.

Krutzikowsky, G. K., & Emmett, R. L. (2005). Diel differences in surface trawl fish catches off Oregon and Washington. *Fisheries Research, 71*, 365–371.

Livingston, P. A. (1983). Food habits of Pacific whiting, *Merluccius productus*, off the west coast of North America, 1967 and 1980. *Fisheries Bulletin U.S., 81*, 629–636.

Livingston, P. A., & Jurado-Molina, J. (2000). A multispecies virtual population analysis of the eastern Bering Sea. *ICES Journal of Marine Science, 57*, 294–299.

Logerwell, E. A., Mantua, N., Lawson, P., Francis, R. C., & Agostini, V. (2003). Tracking environmental processes in the coastal zone for understanding and predicting Oregon coho (*Oncorhynchus kisutch*) marine survival. *Fisheries Oceanography, 12*, 554–568.

MacCall, A. D., Frey, H. W., Huppert, D. D., Knaggs, E. H., McMillan, J. A., & Stauffer, G. D. (1980). Biology and economics of the fishery for jack mackerel in the northeastern Pacific. *NOAA Technical Memorandum, NOAA-TM-NMFS-SWFSC-4*.

MacFarlane, R. G., & Norton, E. C. (2002). Physiological ecology of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) at the southern end of their distribution, the San Francisco Estuary and Gulf of the Farallones, California. *Fisheries Bulletin U.S., 100*, 244–257.

Mantua, N. J., Hare, S. R., Zhang, Y., Wallace, J. M., & Francis, R. C. (1997). A Pacific interdecadal climate oscillation with impacts on salmon production. *Journal of the American Meterological Society, 78*, 1069–1079.

McFarlane, G. A., & King, J. R. (2003). Migration patterns of spiny dogfish (*Squalus acanthias*) in the North Pacific Ocean. *Fisheries Bulletin U.S., 101*, 358–367.

McFarlane, G. A., King, J. R., & Beamish, R. J. (2000). Have there been recent changes in climate? Ask the fish. *Progress in Oceanography, 47*, 147–169.

McGowan, J. A., Cayan, D. R., & Dorman, L. M. (1998). Climate-ocean variability and ecosystem response in the Northeast Pacific. *Science, 281*, 210–217.

Methot, R. D., & Dorn, M. W. (1995). Biology and fisheries of North Pacific hake (*M. productus*). In J. Alheit & T. J. Pitcher (Eds.), *Hake: Biology, fisheries and markets* (pp. 389–414). London: Chapman and Hall.

Murphy, J. M., Brase, A. L. J., & Orsi, J. A. (1999). Survey of juvenile Pacific salmon in the northern region of southeastern Alaska, May–October 1997. *NOAA Technical Memorandum NMFS-AFSC-105*.

Orsi, J. A., Sturdevant, M. V., Murphy, J. M., Mortesen, D. G., & Wing, B. L. (2000). Seasonal habitat use and early marine ecology of juvenile Pacific salmon in southeastern Alaska. *North Pacific Anadromous Fisheries Commission Bulletin, 2*, 111–122.

Pace, M. L., Cole, J. J., Carpenter, S. R., & Kitchell, J. F. (1999). Trophic cascades revealed in diverse ecosystems. *Trends in Ecology and Evolution, 14*, 483–488.

Pearcy, W. G. (1992). *Ocean ecology of North Pacific Salmonids*. Seattle, WA: University of Washington.

Pearcy, W. G., & Fisher, J. P. (1988). Migrations of coho salmon, *Oncorhynchus kisutch*, during their first summer in the ocean. *Fisheries Bulletin U.S., 86*(2), 173–195.

Pearcy, W. G., & Schoener, A. (1987). Changes in the marine biota coincident with the 1982–1983 El Niño in the northeastern subarctic Pacific Ocean. *Journal of Geophysical Research, 92*, 1417–14428.

Peterson, W. T., & Schwing, F. B. (2003). A new climate regime in Northeast Pacific ecosystems. *Geophysical Research Letters, 30*(17), 1896. doi:10.1029/2003GLO157528.

PMFC (Pacific Fishery Management Council) 2005. *Preseason Report I: Stock Abundance Analysis for 2005 Ocean Salmon Fisheries*. Portland, OR: Pacific Fishery Management Council.

PSMFC (Pacific States Marine Fisheries Commission) (2001). *53rd Annual Report of the Pacific States Marine Fisheries Commission*. Portland, OR: Pacific States Marine Fisheries Commission.

Ramsey, F. C., & Schafer, D. W. (1997). *The statistical sleuth*. Belmont, CA: Duxbury Press.

Rexstad, E. A., & Pikitch, E. K. (1986). Stomach contents and food consumption estimates of Pacific hake, *Meluccius productus*. *Fisheries Bulletin U.S., 84*, 947–956.

Schwing, F. B., Bograd, S. J., Collins, C. A., Gaxiola-Castro, G., García, J., Goericke, R., et al. (2002). The state of the California Current, 2001–2002. *California Cooperative Fisheries Investigation Reports, 43*, 31–68.

Shears, N. T., & Babcock, R. C. (2003). Continuing trophic cascade effects after 25 years of no-take marine reserve protection. *Marine Ecology Progressive Series, 246*, 1–16.

Sherwood, C. R., Jay, D. A., Harvey, R. B., Hamilton, P., & Simenstad, C. A. (1990). Historical changes in the Columbia River estuary. *Progress in Oceanography, 25*, 299–352.

Strub, P. T., & James, C. (2003). Altimeter estimates of anomalous transports into the California Current during 2000–2002. *Geophysical Research Letters, 30*(15), 8025, 1–4.

Swain, D. P., & Sinclair, A. F. (2000). Pelagic fishes and the cod recruitment dilemma in the Northwest Atlantic. *Canadian Journal of Fisheries and Aquatic Sciences, 57*, 1321–1325.

Tsou, T. S., & Collie, J. S. (2001). Predation-mediated recruitment in the Georges Bank fish community. *ICES Journal of Marine Science, 58*, 994–1001.

Ware, D. M. (1995). A century and a half of change in the climate of the NE Pacific. *Fisheries Oceanography, 4*, 267–277.

Ware, D. M., & McFarlane, G. A. (1995). Climate-induced changes in Pacific hake (*Merluccius productus*) abundance and pelagic community interactions in the Vancouver Island upwelling system. *Canadian Special Publication of Fisheries and Aquatic Science, 121*, 509–521.

Wheeler, P. A., Huyer, A., & Fleischbein, J. (2003). Cold halocline, increased nutrients and higher chlorophyll off Oregon in 2002. *Geophysical Research Letters, 30*(15), 8021. doi:10.1029/2003GLO17395.

Wiedoff, B., Conrad, J., & Parker, S. (2003). Shoreside hake observation program: 2003. In *Marine Resources Program*. Newport, OR: Oregon Department of Fish and Wildlife.

Williams, J. G., Smith, S. G., Muir, W., Sandford, B. P., Achord, S., McNatt, R., et al. (2005). Effects of the Federal Columbia River power system on salmon populations. *NOAA Technical Memorandum NMFS-NWFSC-63*, 99p.

Zhang, C. I., Lee, J. B., Kim, S., & Oh, J.-H. (2000). Climatic regime shifts and their impacts on marine ecosystem and fisheries resources in Korean waters. *Progress in Oceanography, 47*, 171–190.